[DRAFT] Analyzing lambda using AWS X-Ray

This documentation shows how we can use AWS X-Ray service to perform analysis on lambda, so that we can analyze the latency and total time taken by the lambda.

What is AWS X-Ray:

AWS X-Ray is a distributed tracing service from AWS that helps you track how a request moves through our application and its downstream services.

We use it to:

- see where latency is happening
- to find bottle neck etc.

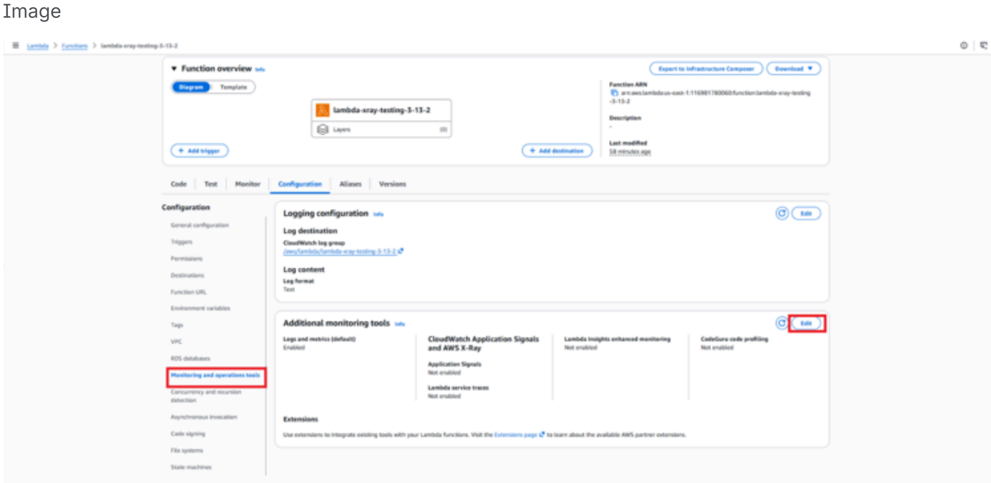
Lambda offers the following monitoring tools

- Application Signals: Application Signals helps you collect application traces and metrics such as requests, availability, latency, errors, and faults
- Lambda service traces: Lambda service X-Ray traces help you track the path of a request through your application.

How to enable for lambda:

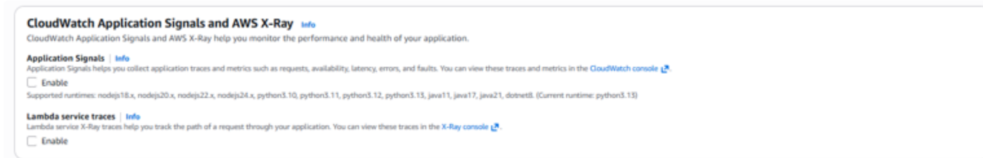
1. Navigate to the lambda you want to enable these traces.
2. Go to `Configuration` tab and select `Monitoring and operations tools`, Click on Edit

Image



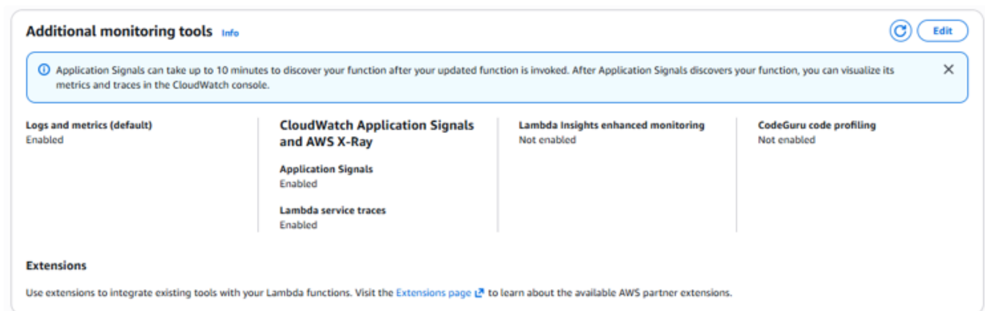
3. Click on `Enable` check box for `Application Signals` and `Lambda service traces` and click on Save.

Image



4. Verify that `Application Signals` and `Lambda service traces` have been enabled.

Image



5. Metric logging have been enabled, you can trigger an event to start analysis.

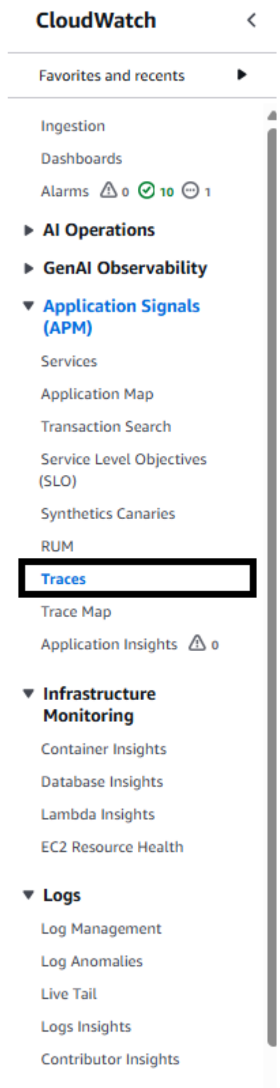
Note:

- To enable `Application Signals`, lambda must me of the following supported versions are `nodejs18.x`, `nodejs20.x`, `nodejs22.x`, `nodejs24.x`, `python3.10`, `python3.11`, `python3.12`, `python3.13`, `java11`, `java17`, `java21`, `dotnet8`.
- Application Signals can take up to 10 minutes to discover your function after your updated function is invoked. After Application Signals discovers your function, you can visualize its metrics and traces in the CloudWatch console.

How to perform analysis:

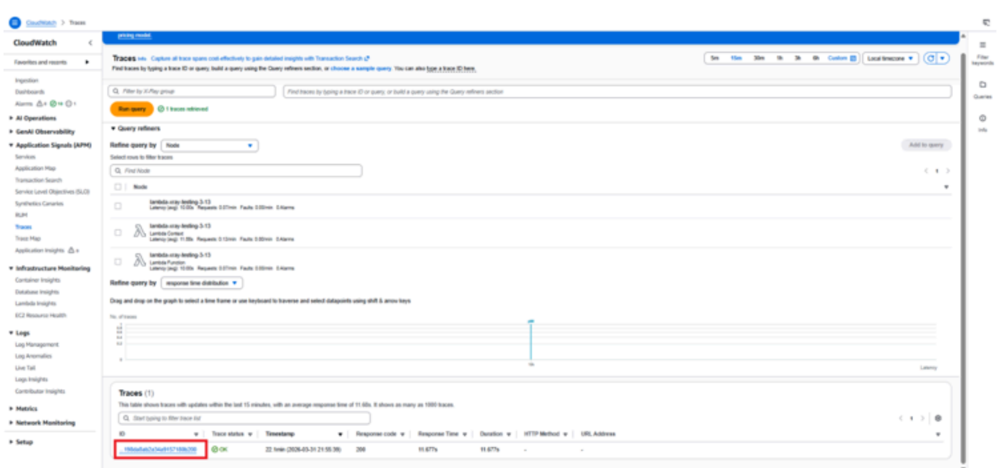
1. To view the traces, go to CloudWatch service, under Application Signals select `Traces`

Image



2. In the `Traces` module, we can see the services we have enabled monitoring, in our case its lambda, in the Traces section select the latest trace (if you have triggered an event recently).

Image



3. After selection we can see the time duration taken for completion of execution and Init time

Image